Preliminary Report

Project 15: *Heart attack prediction*

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1. Task Description

This project consists in implementing a heart attack-predicting model suited for extensive, multi-factor data analysis, framed as a **Binary Classification Problem**.

The model shall learn to ultimately process a set of features and accurately determine whether the patient is **at risk of a heart attack or not**.

2. Dataset Description

2.1 Dataset Choice

As advised in the instructions, the Dataset is being fetched from the *Kaggle* platform. While the platform offers a wide array of *heart attack prediction* related datasets, it is the **UCI Heart Disease** (combined) that has been chosen.

The combined **UCI Heart Disease** dataset aggregates data from Cleveland, Long Beach, Switzerland and Hungary. This dataset is a well-known go-to for such exercises as it provides a multitude of real clinical observations as well as some geographic variance. While the Cleveland subsection of the UCI Heart Disease bundle was the legacy, most complete option, we decided to use the combined version to add some diversity to the overall dataset, increasing the number of records **from 303 to 920**. Finally, the dataset's concise dozen features shall enable a highly relevant feature importance interpretation during the results analysis, compared to other options that might have a greater number of vague, less impactful features.

2.2 Features Breakdown

Feature	Type	Description
age	*numerical*	The age of the patient.
sex	*categorical*	The patient's gender *(male/female)*
cp	*categorical*	Chest Pain diagnosis *(typical/atypical angina, non-anginal, asymptomatic)*
trestbps	*numerical*	Resting blood pressure (mm HG) at admission
chol	*numerical*	Cholesterol level (mg/dl)
fbs	*categorical*	Fasting blood sugar higher than 120mg/dl (*True/False*)
restg	*categorical*	Resting electrocardiographic results (*normal, ST-T abnormality, LV hypertrophy*)
thalach	*numerical*	Maximal heart rate reached during exercise

exang	*categorical*	Exercise-induced angina (*True/False*)
oldpeak	*numerical*	ST depression induced by exercise relative to rest
slope	*categorical*	Slope of peak exercise ST segment
ca	*numerical*	Number of major vessels colored by fluoroscopy (*0 to 3*)
thal	*categorical*	Thalassemia status (*normal, fixed defect, reversible defect*)
num	*numerical*	**Goal variable:** 0 = no disease, 1-4 = increasing risk

> To ensure compliance with the project instructions, the 1-4 `num` values are flattened to 0-1 during preprocessing, 0 remaining 0 while 1, 2, 3 and 4 are converted to 1: there is a **heart attack risk**.

> The dataset also contains the `origin` and `id` features. While the patient ID is pure noise for our model, the impact of a patient's location shall be thoroughly evaluated during the **ablation study** phase, thus determining whether it acts as a **proxy** for unchecked parameters or instead as a **bias**.

3. Proposed Solution

3.1 Model Choice

The **combined UCI Heart Disease** problem can be solved with a wide array of machine-learning models, such as Logistic Regression, K-Nearest Neighbors, Naive Bayes, Support Vector Machines, or ensemble methods like Random Forests or Gradient Boosting.

For this project, **3 models** are being implemented: the `Logistic Regression` (*LR*), `Support Vector Machine` (*SVM*) and `Random Forest Classifier` (*RFC*). While they are all common options for such a task, they also have specific advantages, highly relevant in this context.

Logistic Regression (baseline)	Provides highly interpretable output and not prone to overfitting, thus making the perfect *baseline* for the overall project.
Support Vector Machine	Highly potent for deeply tangled features, but requires precise tuning to avoid overfitting.
Random Forest Classifier	Extremely versatile model and very low overfitting rate. It can however favor certain features and its verdict is unpractical to interpret.

3.2 Data Preprocessing

Once the models are defined, the dataset itself has to be prepared for seamless data processing. The *Combined UCI Heart Disease* dataset is especially challenging as it includes a **significant number of missing fields**, which is something that must be accounted for.

The way *categorical* and *numerical* features are handled differs slightly. Moreover, the preprocessing phase has to be suitable for each model to work with: while the `Random Forest Classifier` would be robust to *missing fields* and *irregular scale numeric values*, these two constraints would respectively crash the `Support Vector Machine` and `Logistic Regression` models and significantly hinder the latters' performances, as they would likely neglect small values such as `ca` (ranging from 0 to 3) for the much higher `chol`.

- **numerical features:**
  - First completing the potential missing values by **Median Imputation**, which effectively fills the voids while being less sensitive to outliers than the *mean* version.
  - Then **normalizing** every value so that no lower yet highly relevant values is neglected. To proceed, the preprocessing phase relies on **Z-score** normalization, achieving an overall *mean value* of 0 and a *standard deviation* of 1.
- **categorical features:**
  - First completing the potential missing values by **implementing a dummy** `Null` **categorical option**, which prevents as mentioned the models from crashing.
  - Then running `OneHotEncoder` to transform worded categorical data into **model-processable matrices**.